Yingrui Hou

Data Science & Scientific Data Engineering

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience turning noisy data into reproducible analyses, checking model reliability and diagnosing performance discrepancies. Built CatBoost workflows, calibrated learned models on real data and used TensorFlow/GPU numerical computing. Applied methods across research problems, coordinated analysis tasks and communicated results internationally. Targeting data science and scientific data engineering.

Core skills

- Model assessment: deep-learning calibration, image-recognition model validation, real-data performance checks, train/test diagnostics and bias correction.
- Modelling: likelihood and mixture models, statistical inference, uncertainty analysis, CatBoost/gradient boosting, feature selection and distribution reweighting.
- Data workflows: reusable processing, Shell/YAML/CMake, Linux, shared-code review and reproducible analysis configurations.
- Programming & computing: Python, NumPy, pandas, scikit-learn, statsmodels; C/C++, ROOT/RooFit; TensorFlow and GPU numerical evaluation.

Research experience

Postdoctoral Researcher | 2021–2026

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Built reusable Python/C++/ROOT workflows for data preparation, weak-signal extraction and statistical validation when individual-event labels were unavailable.
- Trained CatBoost classifiers; checked features, train/test responses and selection thresholds, saving models and configurations for reproducible reuse.
- Aligned simulated, control and target datasets through gradient-boosting reweighting and calibration; propagated residual modelling differences into uncertainty assessments.
- Used real data to assess and optimise an ECAL clustering model; traced reconstruction discrepancies to guide calibration and software follow-up.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

Diagnosing performance discrepancies

Checked an image-recognition model for ECAL clustering against real data. Investigated detector and reconstruction discrepancies and documented findings for calibration and software follow-up. Transferable method: link model evaluation to root-cause investigation.

Calibrating learned-model outputs

Validated and calibrated a deep-learning flavour-tagging model using real data for subsequent analysis. Transferable method: check the reliability of learned-model outputs before using them in a new analysis.

Machine learning under dataset shift

Trained CatBoost classifiers with train/test checks and threshold studies. Reweighted samples to align control and target distributions for subsequent inference. Transferable method: validate model behaviour when samples differ.

Computing complex statistical models

Used TensorFlow and GPU acceleration to construct and evaluate complex-amplitude probability density models in a joint gamma measurement, turning mathematical model definitions into executable numerical computations.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

Learning, delivery & collaboration

- Method transfer: applied statistical inference, CatBoost, TensorFlow/GPU and Geant4 across research tasks, combining literature, implementation and checks on data.
- Delivery: coordinated linked measurement, analysis and validation tasks; built reusable components and documented assumptions for team review.
- Communication: presented at GDR-InF 2022 and FCC/DRD France 2025, including for GRAiNITA; contributed to shared code and technical reviews.

Proposed first assignment: agree the decision and success criteria, check data quality, establish a baseline and deliver a focused validation study with findings and next steps.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- Diagnosing performance discrepancies (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Calibrating learned-model outputs (https://yingrui-hou.github.io/PersonalPage/index.html#work-9)
- Machine learning under dataset shift (https://yingrui-hou.github.io/PersonalPage/index.html#work-3)
- Computing complex statistical models (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)